

Tarea 1: Procesador y Arquitectura

Auditoría y análisis de la CPU en entorno Linux (Ubuntu)

HOST / USUARIO	COMANDO	ARQUITECTURA	ENTORNO
marcooss@Ubuntuu	lscpu	x86_64	VirtualBox / KVM

1. Diagnóstico del Procesador (lscpu)

El comando `lscpu` recolecta información sobre la arquitectura de la CPU, sockets, núcleos, hilos y niveles de memoria caché asignados al sistema operativo.

Modelo de CPU	Intel(R) Core(TM) i7-8750H CPU @ 2.20GHz	Núcleos Asignados	4 Cores (1 socket, 1 hilo/núcleo)
Arquitectura	x86_64 (Modos: 32-bit, 64-bit, Little Endian)	Hipervisor / Tipo	KVM (Virtualización completa)
Jerarquía de Caché	L1d: 128 KiB L1i: 128 KiB L2: 1 MiB L3: 36 MiB		

```
$ lscpu

Architecture:                x86_64
CPU op-mode(s):              32-bit, 64-bit
Address sizes:                39 bits physical, 48 bits virtual
Byte Order:                   Little Endian
CPU(s):                       4
On-line CPU(s) list:         0-3
Vendor ID:                    GenuineIntel
Model name:                   Intel(R) Core(TM) i7-8750H CPU @ 2.20GHz
CPU family:                   6
Model:                        158
Thread(s) per core:           1
Core(s) per socket:           4
Socket(s):                    1
Stepping:                     10
BogoMIPS:                     4416.00
Flags:                        fpu vme de pse tsc msr pae mce cx8 apic sep mtrr pge mca cmov pat
pse36 clflush mmx fxsr sse sse2 ht syscall nx rdtscp lm constant_tsc rep_good nopl xtopology
nonstop_tsc cpuid tsc_known_freq pni pclmulqdq ssse3 fma cx16 pcid sse4_1 sse4_2 x2apic movbe popcnt
aes xsave avx f16c rdrand hypervisor lahf_lm abm 3dnowprefetch pti fsgsbase bmi1 avx2 bmi2 invpcid
rdseed adx clflushopt arat md_clear flush_l1d
Virtualization features:
Hypervisor vendor:            KVM
Virtualization type:           full
Caches (sum of all):
L1d:                           128 KiB (4 instances)
L1i:                           128 KiB (4 instances)
L2:                             1 MiB (4 instances)
L3:                             36 MiB (4 instances)
NUMA:
NUMA node(s):                  1
NUMA node0 CPU(s):            0-3
Vulnerabilities:
Gather data sampling:          Unknown: Dependent on hypervisor status
Ghostwrite:                    Not affected
Indirect target selection:     Mitigation; Aligned branch/return thunks
Itlb multihit:                 KVM: Mitigation: VMX unsupported
L1tf:                          Mitigation; PTE Inversion
Mds:                           Mitigation; Clear CPU buffers; SMT Host state unknown
Meltdown:                      Mitigation; PTI
Mmio stale data:               Mitigation; Clear CPU buffers; SMT Host state unknown
Old microcode:                 Not affected
```

Reg file data sampling:	Not affected
Retbleed:	Vulnerable
Spec rstack overflow:	Not affected
Spec store bypass:	Vulnerable
Spectre v1:	Mitigation; usercopy/swapgs barriers and __user pointer sanitization
Spectre v2:	Mitigation; Retpolines; STIBP disabled; RSB filling; PBRSE-eIBRS Not
affected; BHI Retpoline	
Srbds:	Unknown: Dependent on hypervisor status
Tsa:	Not affected
Tsx async abort:	Not affected
Vmscape:	Not affected